

6th National Contest, BdPhO

C category

Time-3 hours

1st Jan, 2016

Instructions

Make sure your name and other identification tags are written clearly on the script. Also make sure that the answers to the different parts of the questions below are clearly labelled, otherwise BdPhO will not be held responsible for the consequences.

The Exam- 25 marks total

1. Electron on an elliptic track - 10 marks :

The Bohr model for Hydrogen atoms assumes the electron orbits to be circular. This is why it fails to explain the observed fine structure of the atomic spectra hydrogen which was later explained by Arnold Sommerfeld. Sommerfeld considered elliptic orbits for electrons and also took account of relativistic effects. Interestingly, this is an incorrect explanation which was realized only after the development of quantum mechanics. The success of Sommerfeld's work was nothing but an amazing coincidence.

In Sommerfeld's picture the nucleus is assumed to be placed at one of the focus of the elliptical orbits followed by the electrons. The different elliptical orbits are classified by another quantum number k , which was called the azimuthal quantum number -which is similar to the principal quantum number n related to the energy of the orbit. The force between the nucleus and the electron is caused by electrostatic attraction as in Bohr's model.

To facilitate our calculations below, we recommend that one sets the proportionality constant in Coulomb force law k to be equal to one (in other words we replace $\frac{1}{4\pi\epsilon_0} \rightarrow 1$).

- (a) Write the expression for the energy of the electron orbital in terms of the speed v and the separation from the nucleus r .
- (b) If the maximum and minimum distance between the nucleus and the electron (on the same orbit) is r_1 and r_2 respectively, find the ratio of the speeds of these two point $\frac{v_1}{v_2}$.
- (c) Find the expressions of r_1, r_2 in terms of the conserved quantities involved with the orbit.
- (d) Using your result from the above part, show that the energy of the orbit does not depend on the eccentricity of the elliptic orbits.
- (e) If the angular momentum of the orbit is given by $L = k\hbar$ where $\hbar = \frac{h}{2\pi}$ with h being the Planck's constant, express the ratio of semi-minor axis and semi-major axis in terms of the quantum numbers involved with the orbit.

$$1+2+3+2+2=10$$

2. **Passage of Charged Particles through Matter - 9 marks:** Charged particles while passing through matter lose their energy by interacting with the electrons in the medium. The energy loss formula was first derived by Bohr around 1913 and was later given a quantum mechanical treatment by Hans Bethe. In this problem, we will retrace the steps taken for the derivation of a simplified version of the celebrated the Bethe-Bloch formula used for finding the stopping power of different materials. This physics of this process lies at the heart of most detectors used in particle physics.

Consider a massive particle with mass M and charge Ze moving through a medium which has electron density n . The electrons are treated to be free and initially assumed to be at rest. Since the particle is taken to be much more massive compared to electrons, we will assume its path to be a straight line.

- (a) If the perpendicular distance to an electron from the path of the heavy charged particle is b , write the component of the electric force perpendicular to the path of the projectile if its velocity is v and the time passed after passing through the closest point of approach is t .
- (b) Using the above expressions find the change of momentum in the direction perpendicular to the path.

Hint:

- i. you may need to use the integral

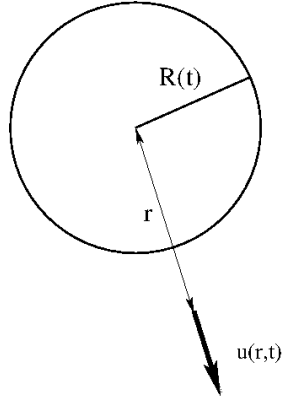
$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^{\frac{3}{2}}} = \frac{2}{a^2}$$

- ii. You can use the Gauss law to derive the same find result for this part.
- (c) Find the energy transferred onto a single electron as a function of the perpendicular distance b .
- (d) Instead of a single electron let us now consider a cloud of electrons, contained within a cylindrical shell whose symmetry axis coincides with the path of heavy charged particle. What is the number of electrons contained in this shell if its inner radius is b , has a shell thickness db and of length dx along the axis?
- (e) Using the above results find the energy loss per path length $\frac{dE}{dx}$, if the electron is spread between a cylindrical sheel of inner radii b_{min} and outer radii b_{max} .
- (f) Your result should diverge (i.e. become infinite) if $b_{min} \rightarrow 0$ and/or $b_{max} \rightarrow \infty$. The minimum value for b_{min} being zero is prevented by Heisenberg uncertainty principle. But suggest how b_{max} can be rendered a finite value to give rise to finite energy loss.

$$1+2 +1+2 +2+1=9$$

3. **Expanding Bubbles- 6 marks:**

Consider a expanding gas bubble which is immersed in an incompressible liquid. Let the radius of the bubble at time t be $R(t)$. We will also assume that the bubble remains spherical during the expansion and its center does not move with time. The liquid outside the bubble undergoes a radial motion due to the expansion of the bubble and let the radial speed of the liquid at a distance r (which is greater than $R(r)$) away from the center be $u(r, t)$ (see the accompanying figure).



- (a) Using the incompressible nature of the liquid find the speed of the liquid $u(r,t)$ as a function of $r, R(t)$ and the expansion speed of the bubble $\frac{dR}{dt} \equiv \dot{R}(t)$.
- (b) The kinetic energy contained in the liquid can be found by integrating the kinetic energy per unit volume of the liquid outside the bubble. Find this kinetic energy in terms of the bubble radius $R(t)$ and the speed of bubble surface $\dot{R}(t)$. [Hint: You will need your result from the previous part]
- (c) This kinetic energy is obtained by the work done by the expansion of the gas. If the pressure of the fluid at $r \rightarrow \infty$ is p_0 , find the relation between the pressure difference and bubble radius and velocity. Choose the initial bubble radius at time $t = 0$ to be R_0 . Your result should be in the form of an integration.
- (d) Differentiating the above result with respect to R and using the relation

$$\frac{\partial \dot{R}^2}{\partial R} = 2\ddot{R}$$

find the differential equation that connects the bubble acceleration and the pressure at the bubble surface. This is a simplified version of the celebrated Rayleigh-Plesset equation.

$$1+2+1+2=6$$